

Choosing a Sampling Strategy Under Evaluation Budget Constraints

Task-completion estimation over a fixed 2,800-session synthetic population under explicit evaluation budget caps.

POPULATION UNITS 2800	AGENT COUNT 105	METHODS 5 (5 expected)	TRIALS 30
CAP RANGE 64-1024	GENERATED AT 2026-08-21 14:21 UTC	MANIFEST STATUS complete	RUN STATUS complete

Decision guide: Lower MAE is better. Mean MAE winner by cap: 64: ARM1 Global Random; 128: ARM1 Global Random; 256: ARM1 Global Random; 512: ARM5 Hajek Weighted; 1024: ARM5 Hajek Weighted. Mean/median disagreement: 1024: mean ARM5 Hajek Weighted, median ARM1 Global Random. Use the recommendation matrix below to balance accuracy against representation and operating constraints.

Executive Summary: Descriptive synthesis only; no inferential significance claims are made.

- Mean MAE winners by cap: ARM1 Global Random: 3 cap winner(s), ARM5 Hajek Weighted: 2 cap winner(s)
- Median MAE winners by cap: ARM1 Global Random: 4 cap median winner(s), ARM5 Hajek Weighted: 1 cap median winner(s)
- Mean/median winner disagreements: cap 1024: mean ARM5 Hajek Weighted, median ARM1 Global Random
- At cap 64, ARM5 mean MAE 0.2318 vs ARM1 0.0417; ARM5 is not broadly competitive at small caps
- Coverage tradeoff: ARM5 vs ARM1 use-case coverage delta: +27.68 pp; classification-derived and provisional because 91.8% of assignments are Ambiguous
- IDW vs selected-only MAE: better in 78/150, worse in 72/150, tied in 0/150 runs
- Token ratio median 4.61%; range 2.42% to 8.93%
- Maven ambiguity caveat: confidence distribution 1:2564, 2:132, 3:81, undetermined:23
- 30-trial descriptive spread now available.

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Purpose and Decision

Audience, decision target, and evidence ladder

Audience: readers new to the project deciding which sampling method to use under explicit budget caps.

Decision: choose the method/cap that best balances aggregate accuracy, representational coverage, and operational cost on the fixed 2,800-session synthetic population.

Aggregate accuracy

Absolute aggregate MAE (lower is better).

Representational coverage

Concept coverage, use-case coverage (classification-derived/provisional), and agent coverage (higher is better).

Operational cost

Actual/Nominal token ratio relative to a 15k planning conversion.

Definitions

- Cap = per-run selected session count target.
- MAE = absolute difference between estimated and census pass rates.
- Concept coverage = represented concept share in the selected sample.
- Use-case coverage = represented Maven use-case share (classification-derived/provisional).
- Agent coverage = represented agent share in selected sessions.

How to read this report

1. Use Executive Recommendations to shortlist method/cap options.
2. Check data provenance and Maven ambiguity caveats.
3. Read question-oriented comparative results and takeaways.
4. Use the Interactive Explorer and sentinel drilldown for deeper checks.

Executive Recommendations

Conditional recommendations by objective and cap

All guidance is descriptive from 30 seeded deterministic replays on a fixed synthetic population; no inferential claims are made.

OBJECTIVE	CAP	RECOMMENDATION	WHY	MAGNITUDE/CAVEAT
Accuracy-first	64/128/256	ARM1	Lowest mean MAE on all three small caps.	ARM1 MAE=0.0417, UC=11.7% ARM5 MAE=0.2318, UC=20.2% (small-cap penalty).
Balanced at mid-high cap	512	ARM5	Best mean MAE at 512 while retaining higher representation than ARM1.	ARM5 MAE=0.0132, UC=82.8% vs ARM1 MAE=0.0157, UC=40.4%.
High-cap near tie	1024	ARM5 (mean) / ARM1 (median)	Near-tie with mean-median disagreement.	Mean MAE favors ARM5, median MAE favors ARM1; treat as operational tie pending shadow run.
Per-agent floor required	Any	ARM3	Only choose when explicit minimum per-agent representation is a hard	ARM3 floor introduces error tradeoff for representation guarantees.

OBJECTIVE	CAP	RECOMMENDATION	WHY	MAGNITUDE/CAVEAT
Imputation study	Any	ARM2	requirement. Use ARM2 for embedding/IDW calibration analysis, not as MAE winner.	IDW improves some runs and worsens others; not the aggregate MAE leader.

Small-cap penalty and crossover: ARM5 mean MAE at cap64 is 0.2318 vs ARM1 0.0417. At cap1024, ARM5 wins mean MAE (0.0063 vs 0.0070) while ARM1 wins median MAE (0.0054 vs 0.0058).

Dataset and Ground Truth

Two corpora, normalization, and bounded evidence examples

dense_2500: 2,500 sessions, 5 named agents (500 each), seed 3652026, scenario/difficulty/variety/behavior flags with templated tool calls and pass/fail narration; outputs/manifests retained while generator source is absent from this workspace.

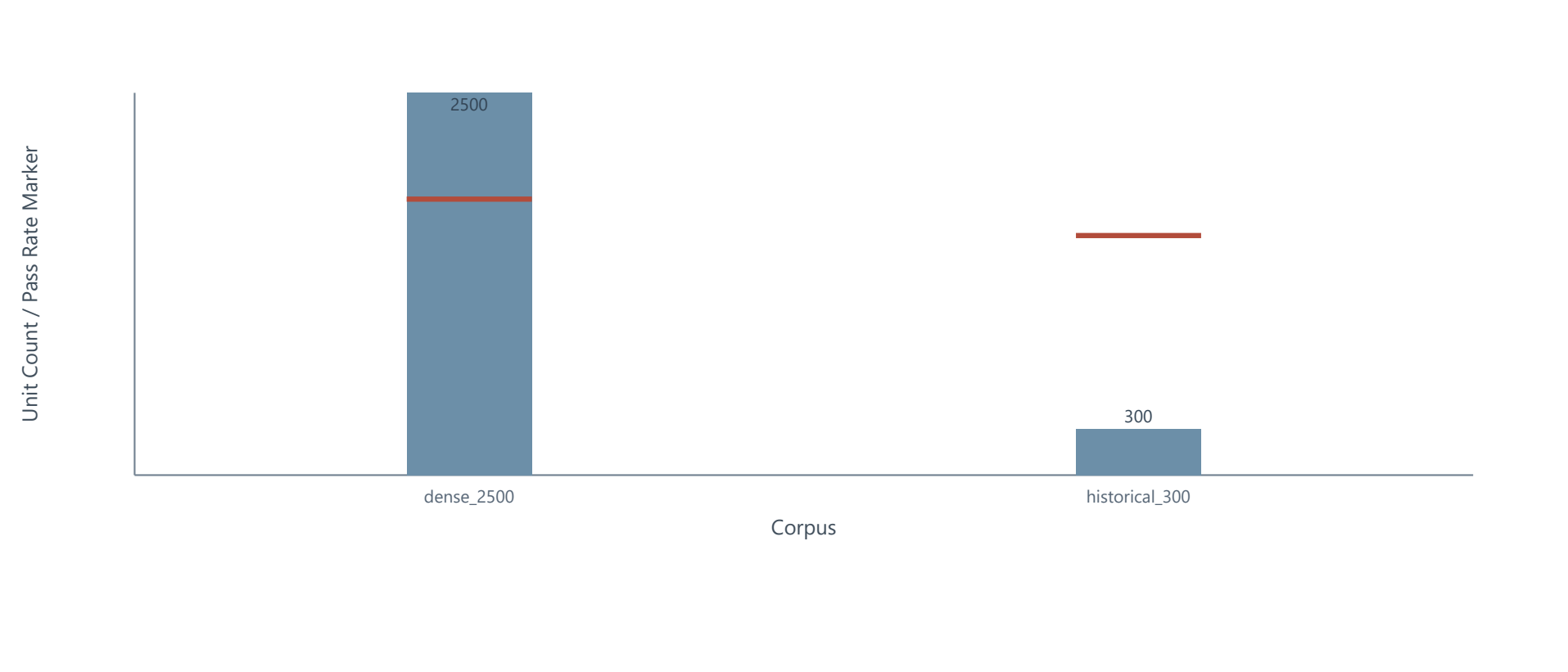
historical_300: 300 sessions, 100 agents from BPS converted by synthetic_data/a365_historical_300/build_a365_otel.py; pass/fail copied from upstream expected field with deterministic IDs/timestamps where needed; original BPS authoring is outside this workspace.

Combined census: 2800 sessions, 105 agents, pass rate 71.1786%. Source pass rates: dense_2500 72.2000% and historical_300 62.6667%.

OTLP-style traces are normalized into EvaluationUnit rows (unit IDs, turns, tools); expected labels are joined only after sample membership is frozen. Source-synthetic fields are distinct from report-derived snippets.

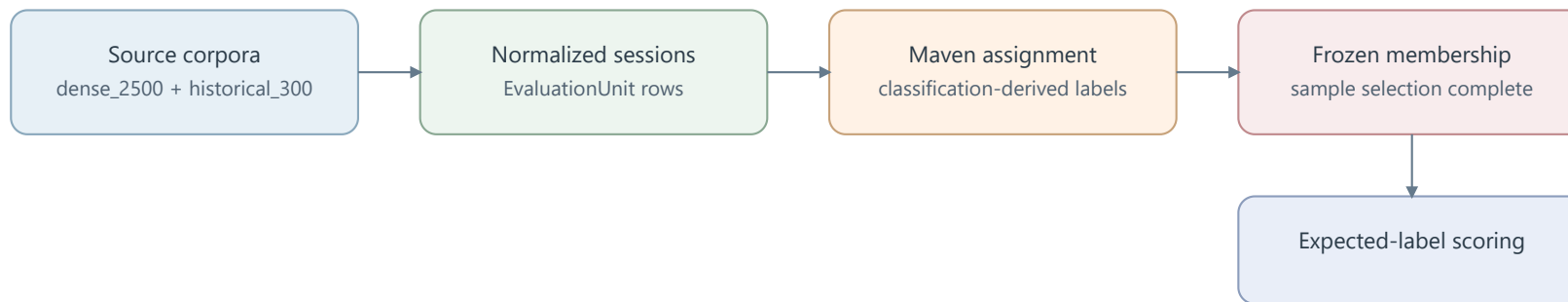
Source Composition: Counts and Pass Rates

Bars are unit counts; red markers are source pass rates.



Data Pipeline Overview

Source corpora to scoring with frozen memberships.



Example 1: corpus dense_2500

Agent

tenant-a365-deterministic|procurement-agent

Expected Label

True

Source Metadata

synthetic=yes; hash=76da6dc93ce0bb485117fb99

Turn / Tool / Error Shape

turns=1; tools=4; had_error=no

Metadata

domain=vendor-management; task=rfp_screening; pass_rate=N/A

User Snippet (bounded preview)

Task rfp_screening in vendor-management (renewal-cycle). Need policy-safe recommendation for fictional business request ...

Assistant Snippet (bounded preview)

Outcome pass: request completed with compliant recommendation and clear action steps. Borderline confidence remains beca...

Example 2: corpus dense_2500**Agent**

tenant-a365-deterministic|procurement-agent

Expected Label

unknown

Source Metadata

synthetic=yes; hash=76da6dc93ce0bb485117fb99

Turn / Tool / Error Shape

turns=1; tools=7; had_error=no

Metadata

domain=vendor-management; task=rfp_screening; pass_rate=N/A

User Snippet (bounded preview)

Task rfp_screening in vendor-management (new-vendor-onboarding). Need policy-safe recommendation for fictional business ...

Assistant Snippet (bounded preview)

Outcome fail: unresolved blocker remains; assistant provides bounded fallback and follow-up path.

Example 3: corpus historical_300

Agent

badf1f56-284d-4dc5-ac59-0dd53900e743|listing-advisor-agent

Expected Label

True

Source Metadata

synthetic=yes; hash=b1afbc395bfa022d6a5b08a0

Turn / Tool / Error Shape

turns=3; tools=3; had_error=no

Metadata

domain=real-estate; task=[domain: real-estate] suggest list price; draft MLS description; schedule open house; pass_rate=N/A

User Snippet (bounded preview)

I'm listing a 3-bed, 2-bath single-family home at 142 Elm St, 1,850 sqft. What list price should I suggest?

Assistant Snippet (bounded preview)

Based on 6 recent comparable sales nearby, I'd suggest listing 142 Elm St in the \$415K-\$455K range (median ~\$432K). That...

Example 4: corpus historical_300

Agent

badf1f56-284d-4dc5-ac59-0dd53900e743|test-case-generator

Expected Label

True

Source Metadata

synthetic=yes; hash=b1afbc395bfa022d6a5b08a0

Turn / Tool / Error Shape

turns=3; tools=4; had_error=no

Metadata

domain=qa-testing; task=[domain: qa-testing] generate functional + edge test cases for REQ-2291 and create the suite;
pass_rate=N/A

User Snippet (bounded preview)

Generate functional test cases for REQ-2291 (email + OTP login).

Assistant Snippet (bounded preview)

Pulled REQ-2291 "Email + OTP login" (acceptance: valid OTP logs in, expired OTP rejected, 3 failed attempts locks 15m). ...

What to notice: four bounded case-study examples cover pass/fail behavior across both corpora and show lineage + shape metadata without exposing full transcripts.

Source Summary + Fidelity

Overall units: 2800; pass count: 1993; pass rate: 71.18%

Source hashes, corpus counts, and pass-rates are report-derived from the run artifact. Snippets shown below are bounded previews and are not full transcripts.

CORPUS	UNITS	PASS COUNT	PASS RATE	SOURCE HASH PREFIX
dense_2500	2500	1805	72.20%	76da6dc93ce0bb485117fb99
historical_300	300	188	62.67%	b1afbc395bfa022d6a5b08a0

What Is Synthetic vs Report-Derived

- description: Combined synthetic evaluation corpus built from source corpora and normalized into EvaluationUnit rows.
- expected_label_field: labels_by_unit (boolean pass/fail)
- snippet_policy: bounded first-turn preview from source text

source_synthetic

- expected_label
- shape.turn_count
- shape.tool_call_count

- shape.had_error

report_derived

- snippet.user
- snippet.assistant

Maven Business Use-Case Assignment

Workflow diagnostics, thresholds, and provisional coverage caveat

Workflow: cleaning, per-turn request/response embeddings with text-embedding-3-small, real v6 request/response centroids, top-5 agreement/corroboration/late-fusion, 0.30 low-confidence sentinel, 0.70 session early-stop, otherwise maximum similarity assignment.

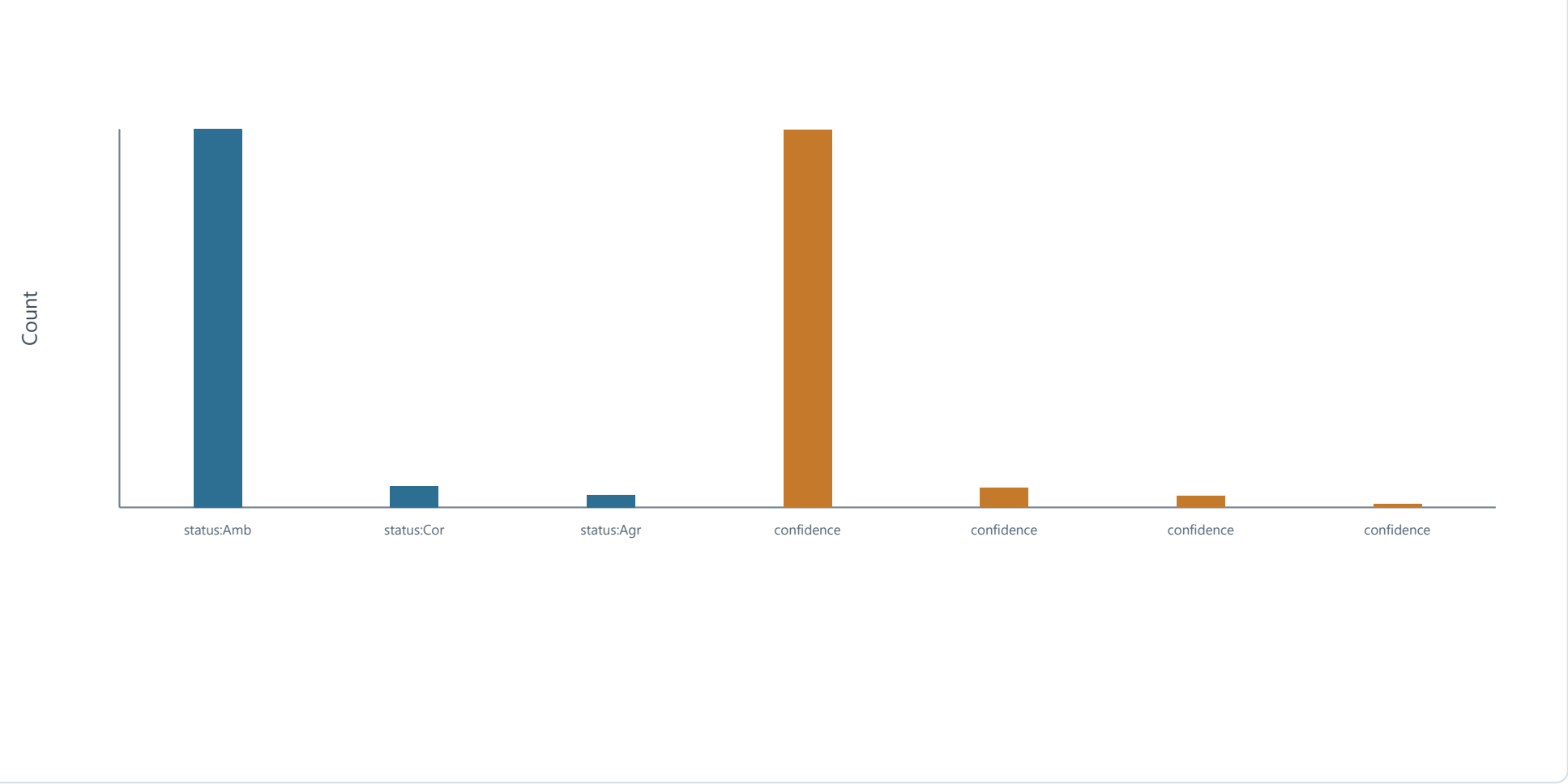
This section describes classification behavior diagnostics, not an accuracy benchmark or external validation of the assigned use cases.

Observed bundle: Ambiguous 2569 (91.8%), Corroborated 146, Agree 85; confidence level 1=2564; Below-0.30 fallback=23; max similarity 0.690. Use-case coverage is therefore provisional.

Per-corpus diagnostics: dense_2500 n=2500, mean sim 0.527, status [('Ambiguous', 2486), ('Corroborated', 14)]. historical_300 n=300, mean sim 0.499, status [('Corroborated', 132), ('Agree', 85), ('Ambiguous', 83)], confidence [('2', 118), ('3', 81), ('1', 78), ('undetermined', 23)].

Maven Status + Confidence Distribution

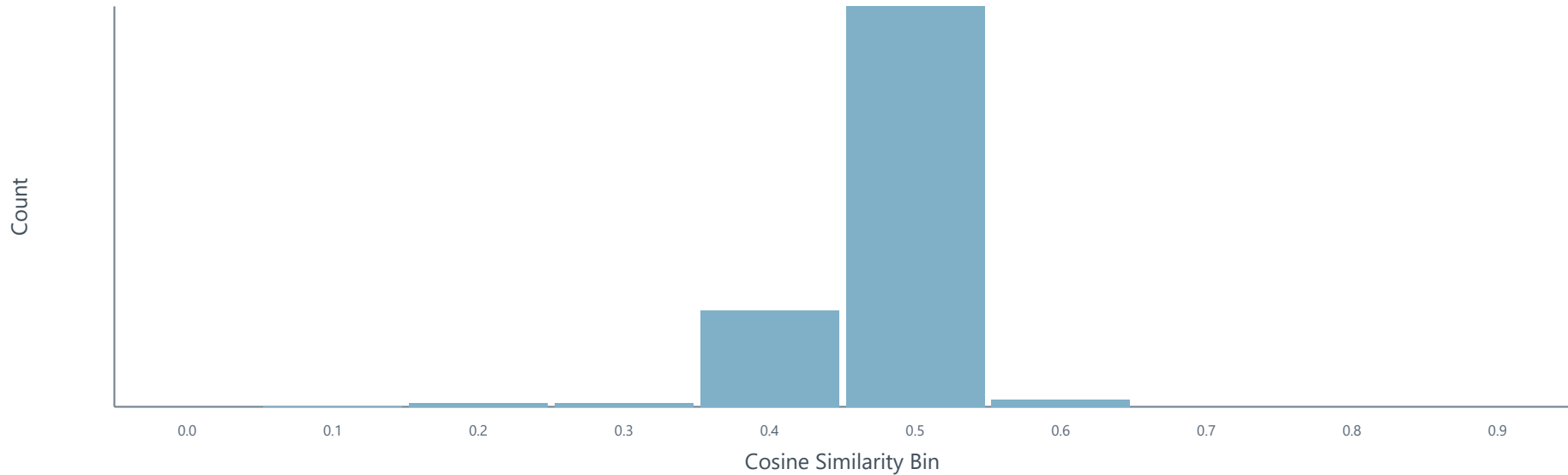
Bars summarize classifier output status and confidence buckets.



Takeaway: Status mix is dominated by Ambiguous, so use-case coverage remains classification-derived/provisional.

Maven Similarity Histogram

Histogram over combined cosine similarity scores.



What this shows: Similarity values cluster below the 0.70 early-stop threshold; 0.30 and 0.70 are operational cutoff annotations for weak fallback and strong assignment.

Top Use Cases (Human-readable labels, GUID secondary)

- **Financial Compliance Review / Financial compliance review** (fc8e4c9e-d911-4fa8-8719-708f9b2b9651): 260
- **Travel Policy Compliance / Travel policy compliance** (2c698729-251a-4ebe-aec5-c81e2be1dce9): 218
- **Incomplete or Vague Prompt / Vague prompt clarification** (0231ac82-2e61-4c02-907e-17ee61eefd56): 136

- **Issue Resolution / Issue resolution** (51f99f1a-4357-4cf4-9777-c353fc24d48d): 125
- **Dependency Risk / Dependency risk assessment** (61d98c2f-1954-4e3d-95cf-d1f0c79fab59): 117
- **Rollback Procedures / Rollback procedure development** (ab3f7996-8a78-43e5-b3e3-bbba78b2667d): 113
- **Third-Party Risk / Third-party risk assessment** (4f387cbf-1796-4201-8c45-99dde6309f87): 104
- **Compliance Screening / Compliance screening** (d4fc97da-3222-4678-a9e5-b075087da334): 85
- **Graceful Degradation / Graceful degradation design** (6ce3bd74-e441-4886-b247-a0e4befabbcd): 64
- **Service Request Handling / Service request handling** (fd2ccf35-9332-4cb7-85f7-7857e943a4e8): 56
- **Corrective Action Tracking / Corrective action tracking** (54f9eeea-baa6-4fef-97bb-16ad2c793bf2): 55
- **Edge Cases / Edge case identification** (a032f981-8e5f-48c1-b85e-2fc43ef08260): 53

Compact taxonomy mapping table

DOMAIN	CATEGORY	SUB-SUBCATEGORY	BUSINESS TASK	COUNT
Government & Public Sector	Public Finance & Taxation	Financial Compliance Review	Financial compliance review	260
Administration & Workplace Services	Executive & Administrative Support	Travel Policy Compliance	Travel policy compliance	218
Meeting, Productivity & Assistant Interaction	Ambiguous & Underspecified	Incomplete or Vague Prompt	Vague prompt clarification	136
Customer Service & Success	Field & Technical Service	Issue Resolution	Issue resolution	125
Cross-Functional Capabilities	Program & Project Management	Dependency Risk	Dependency risk assessment	117
Information Technology & Infrastructure	IT Service Management	Rollback Procedures	Rollback procedure development	113
Legal, Risk & Compliance	Risk Management	Third-Party Risk	Third-party risk assessment	104
Procurement & Vendor Management	Supplier Relationship & Risk	Compliance Screening	Compliance screening	85
Software & Application Engineering	Distributed Systems & Microservices	Graceful Degradation	Graceful degradation design	64
Banking, Insurance & Financial Services	Customer Experience & Marketing	Service Request Handling	Service request handling	56
Information Technology & Infrastructure	Monitoring & Reliability	Corrective Action Tracking	Corrective action tracking	55
Product & Design	Product Management	Edge Cases	Edge case identification	53
Manufacturing & Production	Product Lifecycle & Engineering	Lifecycle Stage Gate Review	Lifecycle stage gate review	52
Travel & Hospitality	Travel Services & OTA	24/7 Traveler Assistance	24/7 Traveler assistance	50
Legal, Risk & Compliance	Audit & Assurance	Corrective Action Plans	Corrective action planning	46

Experiment Design and Metrics

Fixed population design, replay pairing, and metric intent

Design: fixed population, caps 64/128/256/512/1024, 30 paired deterministic seeds, and membership frozen before expected-label joins. Estimand is pass-rate estimation quality and representation behavior on this fixed population.

Descriptive spread is reported as mean/median/p05/p95; these are not inferential confidence intervals.

15k is a planning-only conversion from token budget to session cap; observed actual/nominal ratios (2.42% to 8.93%) are expected and do not imply selection-budget under-run.

Metric glossary

- Absolute aggregate MAE (lower better): estimator accuracy.
- Concept coverage (higher better): source concept representation.
- Use-case coverage (higher better, provisional): classification-derived representation.
- Agent coverage (higher better): cross-agent representation.
- Actual/Nominal token ratio: operational cost context.

Method Guide

Selector/estimator matrix and method-specific caveats

METHOD	SELECTOR	ESTIMATOR	INTENDED BENEFIT	MAIN RISK
ARM1	Global deterministic random	Unweighted mean	Best low-cap aggregate accuracy baseline	Can miss minority agents/use-cases at small caps
ARM2	Azure Search adaptive selector over full-session embeddings	Same-agent IDW over full population	Model-assisted estimation with donor reuse	Donor/fallback chain can amplify calibration drift
ARM3	One-per-agent round-robin to min(3, N_agent), then Maven strata	Unweighted mean	Explicit per-agent floor representation	Floor pressure can hurt MAE under tight caps
ARM4	Capacity-aware per-agent round robin + proportional use-case strata	Unweighted mean	Coverage-focused membership without hard floor	No formal minimum per-agent guarantee
ARM5	Exact ARM4 membership	Hajek ratio estimator with inclusion probabilities	Same coverage as ARM4 with weighted estimator	Low-cap weighting instability; small-cap MAE penalty

ARM1 global deterministic random + unweighted mean

Strong low-cap MAE baseline with simpler behavior.

ARM2 adaptive selector + same-agent IDW

Full-session embeddings and Azure Search donor retrieval with fallback chain; primary use is imputation/calibration study.

ARM3 floor then strata + unweighted mean

One-per-agent round robin to $\min(3, N_{\text{agent}})$, then proportional Maven strata. The artifact-derived floor target totals 163 sessions.

ARM4 capacity-aware round robin + unweighted mean

No hard floor; emphasizes broad representation via round-robin and strata.

ARM5 ARM4-membership + Hajek ratio estimator

Coverage mirrors ARM4; only estimator differs. Can be unstable at low caps.

Token-cap conversion and equations:

Generated Methodology Narrative (Artifact-Driven)

This narrative is synthesized from artifact fields and does not depend on optional markdown richness.

Population and replay setup: caps=64, 128, 256, 512, 1024; observed trial count=30. A 30-trial spread is available in this bundle.

1. Population and labels: full-session packet units are labeled with expected outcomes and attached metadata.
2. Five-arm design: ARM1 global random, ARM2 embedding-IDW, ARM3 round-robin with floor, ARM4 round-robin strata balancing, ARM5 same-membership Hajek weighted estimator.
3. Cap conversion: sample budget uses a 15k tokens/session planning conversion.
4. Maven classifier: confidence thresholds 0.30 and 0.70 partition ambiguous vs stronger assignments.

5. Embedding and clustering: full-session packet embeddings are grouped through Azure Search neighborhood retrieval for donor selection.
6. IDW donor chain: observed value, exact/donor interpolation, then agent/global/prior fallback groups when donor evidence is sparse.
7. ARM3 floor then strata: minimum floor target first (configured min-per-agent=3), then remaining budget allocated by strata.
8. ARM4 stratification: round-robin strata exposure without hard floor completion requirement.
9. ARM5 same-membership Hajek estimator: uses membership-aligned weighted ratio estimator for aggregate rates.
10. Metrics and spread: MAE, concept/use-case/agent coverage, token ratio, Brier and ECE; trial spread reported as min/max descriptive ranges.
11. Trial pairing and seeds: runs are paired by cap and seed; spread is descriptive and not inferential significance testing.
12. Limitations: classifier ambiguity, sparse donor contexts, and cap-limited coverage can bias observed rates.

Core Equations

Cap from token budget: $n = \left\lfloor \frac{B}{15000} \right\rfloor$

Absolute aggregate MAE: $MAE = \frac{1}{K} \sum_{k=1}^K | \hat{r}_k - r_k |$

Concept coverage: $C_{concept} = \frac{|U_{selected \cap concept}|}{|U_{concept}|}$

IDW estimate: $\hat{y}(x) = \frac{\sum_i^m w_i(x) y_i}{\sum_i^m w_i(x)} \quad w_i = \frac{1}{d^p}$

Hajek estimator: $\hat{RH} = \frac{\sum_i^n \frac{y_i}{\pi_i}}{\sum_i^n \frac{1}{\pi_i}}$

Source Methodology (Sanitized Markdown)

Sampling V6 Methodology

Scope

- Population is the exact combined 2800-session retained synthetic corpus unless canary slicing is explicitly requested.
- Expected labels are joined after membership freeze for every arm.

Arms

- Arm1: global deterministic random membership.
- Arm2: embedding-based adaptive selection replayed against the real vector index with exact-count cap and IDW estimation over the full population.
- Arm3: agent round-robin with floor prefix.
- Arm4: agent round-robin probability design.
- Arm5: Hajek-weighted reuse of Arm4 membership and inclusion probabilities.

Arm2 Exact-Count Logic

- Replays a deterministic seed order through AdaptiveSampler + AzureClusterIndex via V3EmbeddingSelector.
- Native proposals are accepted in replay order if `native_count <= cap`.
- If `native_count > cap`, native rows are ranked by novelty desc, rarity desc, stable hash and truncated to cap.
- If `native_count < cap`, fill rows are ranked by novelty desc, rarity desc, stable hash and appended until exact cap.
- Any fallback-signature behavior causes hard failure.
- Azure scope is cleaned before and after each arm2 replay; settled zero remaining docs is required when the store supports scope counting.

Token Accounting

- Nominal budget per run is $\text{cap} * \text{avg_tokens_per_session}$.
- Actual token count is the sum of runtime token costs for selected unit IDs.

Maven Classification Parity

- Each turn is mapped in order to `SessionStepText(request=user_text, response=assistant_text)`.
- Cleaning and token limits use classifier-native logic and `cl100k_base` counting.
- Classification applies threshold early-stop then max-similarity fallback exactly as classifier implementation defines.
- classification artifacts store no raw session text.

Estimands

- Aggregate estimand is pass-rate over full population.

- Arm2 uses observed+IDW imputation and reports validation diagnostics.
- Arm5 uses Hajek ratio estimator with Arm4 inclusion probabilities.

Limitations

- Arm2 exact-count replay introduces deterministic ranking choices when proposals exceed cap.
- IDW is model-assisted and should be treated as an approximation, not a design-unbiased guarantee.
- Existing dataset metadata and expected labels are synthetic and non-production.

Worked examples

Worked Example: ARM2 IDW Row

Cap=64, seed=13, observed=64, imputed=2736.

Selected-only rate/error: 68.75% / 0.0243. IDW estimated rate/error: 75.59% / 0.0442.

Worked Example: ARM3 Membership Floor

Cap=64, floor target=163, prefix selected=64, completion=39.26%.

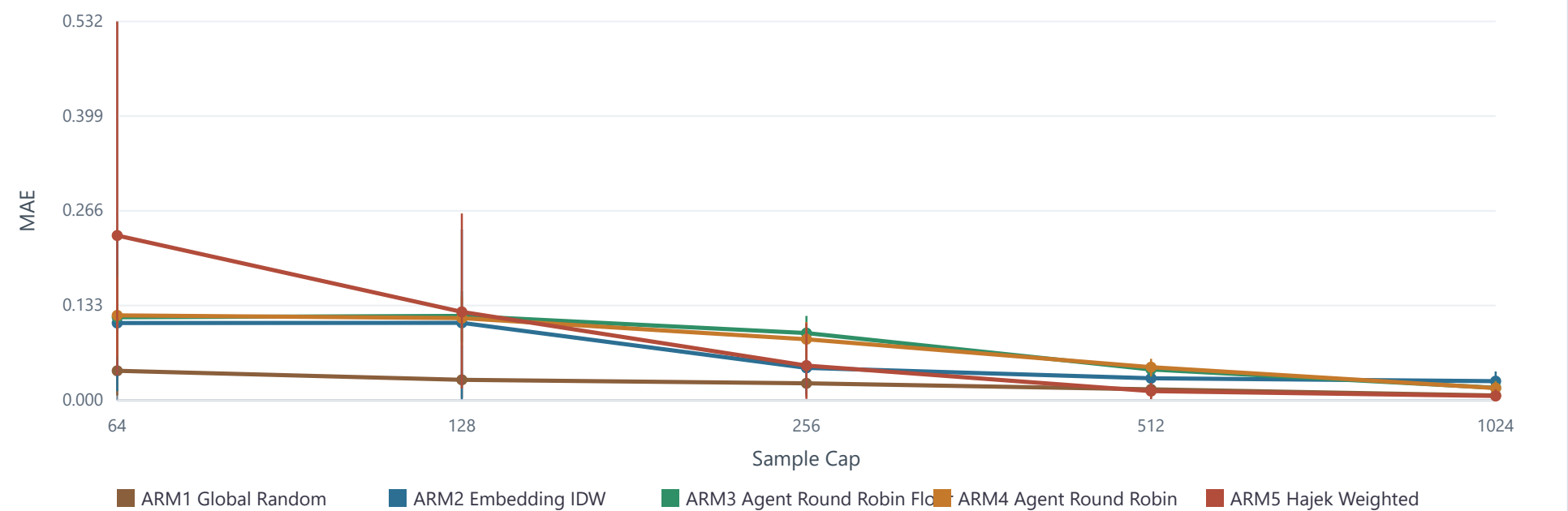
Comparative Results

Question-oriented evidence with explicit takeaways

a. Accuracy by cap

MAE vs Cap with Trial Spread

Lines show per-method mean with empirical 5th-95th percentile descriptive spread for n>=10; smaller bundles use min/max.



Takeaway: Lower MAE is better. Mean MAE winner by cap: 64: ARM1 Global Random; 128: ARM1 Global Random; 256: ARM1 Global Random; 512: ARM5 Hajek Weighted; 1024: ARM5 Hajek Weighted. Mean/median disagreement: 1024: mean ARM5 Hajek Weighted, median ARM1 Global Random.

METHOD	CAP	MAE	CONCEPT COVERAGE	USE-CASE COVERAGE	AGENT COVERAGE
ARM1 Global Random	64	0.0417	11.79%	11.68%	9.56%
ARM2 Embedding IDW	64	0.1088	13.33%	18.62%	44.67%
ARM3 Agent Round Robin Floor	64	0.1169	13.33%	20.15%	60.95%
ARM4 Agent Round Robin	64	0.1196	13.33%	20.17%	60.95%

METHOD	CAP	MAE	CONCEPT COVERAGE	USE-CASE COVERAGE	AGENT COVERAGE
ARM5 Hajek Weighted	64	0.2318	13.33%	20.17%	60.95%
ARM1 Global Random	128	0.0290	20.84%	18.77%	14.98%
ARM2 Embedding IDW	128	0.1091	26.67%	35.03%	76.76%
ARM3 Agent Round Robin Floor	128	0.1188	26.64%	37.95%	100.00%
ARM4 Agent Round Robin	128	0.1157	26.67%	38.50%	100.00%
ARM5 Hajek Weighted	128	0.1245	26.67%	38.50%	100.00%
ARM1 Global Random	256	0.0241	33.74%	27.71%	21.78%
ARM2 Embedding IDW	256	0.0458	51.38%	58.33%	100.00%
ARM3 Agent Round Robin Floor	256	0.0947	52.15%	59.41%	100.00%
ARM4 Agent Round Robin	256	0.0860	52.36%	60.40%	100.00%
ARM5 Hajek Weighted	256	0.0490	52.36%	60.40%	100.00%
ARM1 Global Random	512	0.0157	46.80%	40.40%	33.17%
ARM2 Embedding IDW	512	0.0312	73.97%	71.44%	100.00%
ARM3 Agent Round Robin Floor	512	0.0436	88.60%	82.24%	100.00%
ARM4 Agent Round Robin	512	0.0468	88.92%	82.75%	100.00%
ARM5 Hajek Weighted	512	0.0132	88.92%	82.75%	100.00%
ARM1 Global Random	1024	0.0070	60.33%	57.00%	50.95%
ARM2 Embedding IDW	1024	0.0271	83.15%	81.04%	100.00%
ARM3 Agent Round Robin Floor	1024	0.0179	99.71%	92.15%	100.00%
ARM4 Agent Round Robin	1024	0.0174	99.75%	92.13%	100.00%
ARM5 Hajek Weighted	1024	0.0063	99.75%	92.13%	100.00%

b. Representation by cap

Coverage vs Cap: Source Concept

Lines show per-method mean with empirical 5th-95th percentile descriptive spread for $n \geq 10$; smaller bundles use min/max.



Takeaway: Concept coverage generally rises with cap and is stronger for representation-focused methods.

Coverage vs Cap: Maven Use-Case

Lines show per-method mean with empirical 5th-95th percentile descriptive spread for $n \geq 10$; smaller bundles use min/max.



Takeaway: Use-case coverage trends are informative but classification-derived/provisional because 91.8% of assignments are Ambiguous.

Coverage vs Cap: Agent Coverage

Lines show per-method mean with empirical 5th-95th percentile descriptive spread for $n \geq 10$; smaller bundles use min/max.

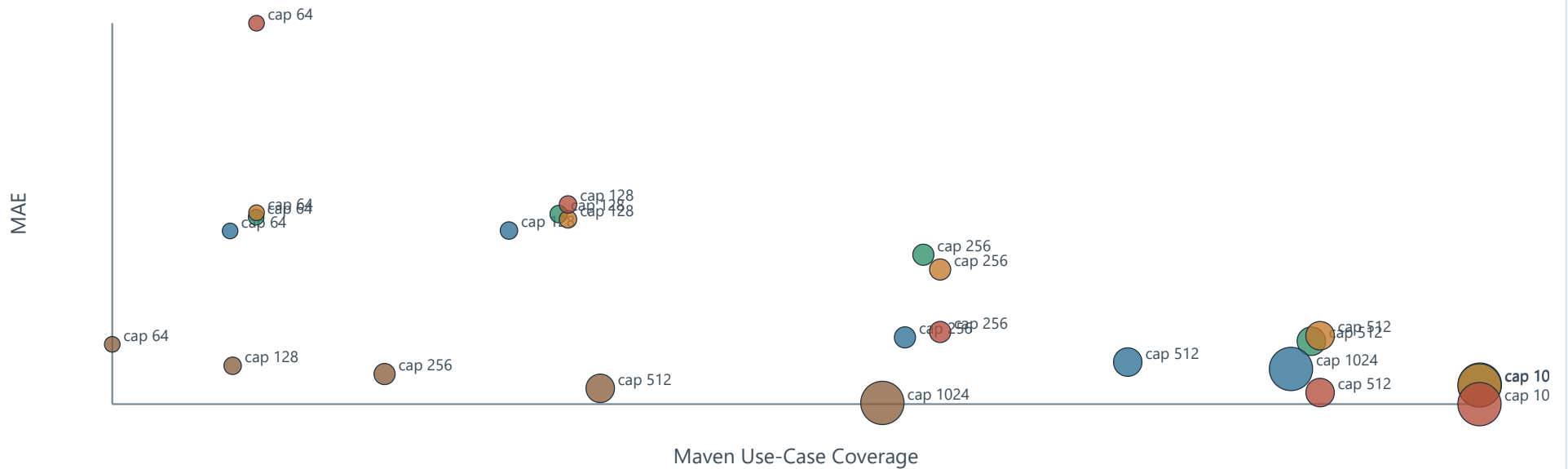


Takeaway: Agent coverage increases with cap; floor/round-robin designs prioritize broader agent representation.

c. Accuracy-coverage frontier

MAE vs Maven Use-Case Coverage Frontier

Higher coverage (x-axis) and lower MAE (y-axis) is preferable. Point size scales with cap.



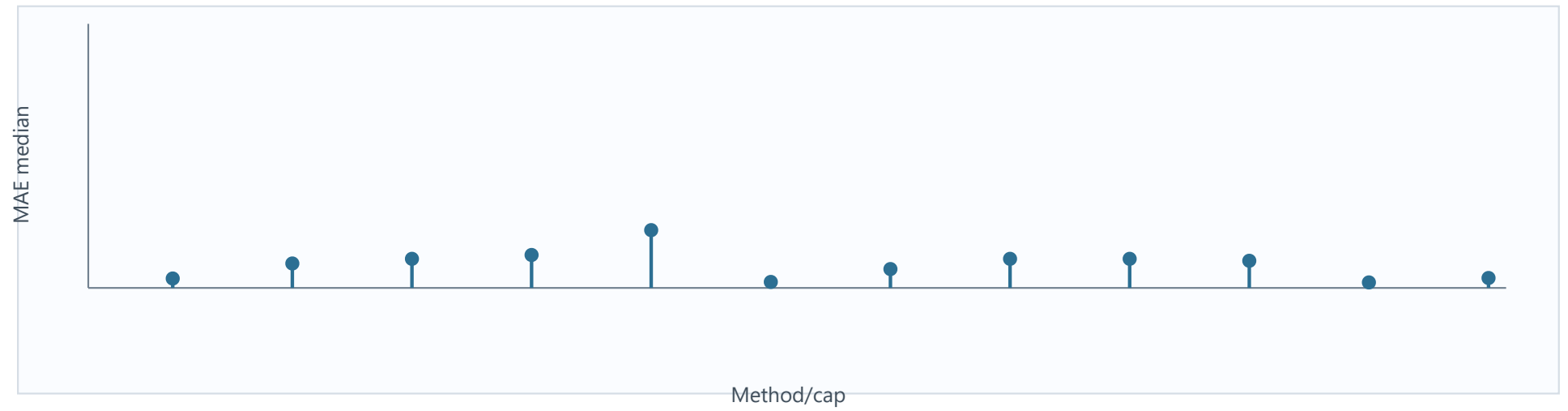
METHOD	CAP	MAE MEAN	MEDIAN	SD	SPREAD	WIN RATE VS ARM1
ARM1 Global Random	64	0.0417	0.0382	0.0313	0.0070–0.1023	N/A
ARM1 Global Random	128	0.0290	0.0243	0.0225	0.0036–0.0762	N/A
ARM1 Global Random	256	0.0241	0.0223	0.0159	0.0048–0.0559	N/A
ARM1 Global Random	512	0.0157	0.0146	0.0101	0.0028–0.0315	N/A
ARM1 Global Random	1024	0.0070	0.0054	0.0064	0.0008–0.0182	N/A
ARM2 Embedding IDW	64	0.1088	0.0992	0.0775	0.0131–0.2393	23.3%
ARM2 Embedding IDW	128	0.1091	0.0765	0.0815	0.0017–0.2402	20.0%
ARM2 Embedding IDW	256	0.0458	0.0403	0.0284	0.0090–0.0978	23.3%
ARM2 Embedding IDW	512	0.0312	0.0282	0.0133	0.0125–0.0520	23.3%
ARM2 Embedding IDW	1024	0.0271	0.0265	0.0088	0.0175–0.0407	3.3%
ARM3 Agent Round Robin Floor	64	0.1169	0.1180	0.0495	0.0555–0.1962	10.0%
ARM3 Agent Round Robin Floor	128	0.1188	0.1180	0.0240	0.0817–0.1536	0.0%
ARM3 Agent Round Robin Floor	256	0.0947	0.0966	0.0129	0.0768–0.1188	0.0%
ARM3 Agent Round Robin Floor	512	0.0436	0.0438	0.0079	0.0330–0.0547	0.0%
ARM3 Agent Round Robin Floor	1024	0.0179	0.0174	0.0063	0.0096–0.0278	10.0%
ARM4 Agent Round Robin	64	0.1196	0.1337	0.0368	0.0626–0.1735	10.0%
ARM4 Agent Round Robin	128	0.1157	0.1180	0.0255	0.0747–0.1458	0.0%
ARM4 Agent Round Robin	256	0.0860	0.0848	0.0194	0.0594–0.1102	0.0%
ARM4 Agent Round Robin	512	0.0468	0.0487	0.0080	0.0341–0.0586	3.3%
ARM4 Agent Round Robin	1024	0.0174	0.0184	0.0066	0.0057–0.0269	16.7%
ARM5 Hajek Weighted	64	0.2318	0.2345	0.1428	0.0354–0.5325	13.3%
ARM5 Hajek Weighted	128	0.1245	0.1101	0.0874	0.0170–0.2627	13.3%
ARM5 Hajek Weighted	256	0.0490	0.0400	0.0406	0.0022–0.1088	30.0%
ARM5 Hajek Weighted	512	0.0132	0.0115	0.0071	0.0019–0.0248	53.3%
ARM5 Hajek Weighted	1024	0.0063	0.0058	0.0052	0.0003–0.0158	60.0%

Paired comparisons

COMPARISON	CAP	MAE DELTA	P05–P95	WIN RATE	ESTIMATE DELTA
ARM2 Embedding IDW vs ARM1	64	0.0671	-0.0442–0.2030	23.3%	N/A
ARM2 Embedding IDW vs ARM1	128	0.0800	-0.0224–0.2202	20.0%	N/A
ARM2 Embedding IDW vs ARM1	256	0.0217	-0.0139–0.0801	23.3%	N/A

COMPARISON	CAP	MAE DELTA	P05-P95	WIN RATE	ESTIMATE DELTA
ARM2 Embedding IDW vs ARM1	512	0.0155	-0.0072–0.0489	23.3%	N/A
ARM2 Embedding IDW vs ARM1	1024	0.0201	0.0046–0.0362	3.3%	N/A
ARM3 Agent Round Robin Floor vs ARM1	64	0.0752	-0.0408–0.1736	10.0%	N/A
ARM3 Agent Round Robin Floor vs ARM1	128	0.0898	0.0348–0.1308	0.0%	N/A
ARM3 Agent Round Robin Floor vs ARM1	256	0.0706	0.0391–0.1001	0.0%	N/A
ARM3 Agent Round Robin Floor vs ARM1	512	0.0279	0.0066–0.0460	0.0%	N/A
ARM3 Agent Round Robin Floor vs ARM1	1024	0.0108	-0.0060–0.0221	10.0%	N/A
ARM4 Agent Round Robin vs ARM1	64	0.0779	-0.0149–0.1353	10.0%	N/A
ARM4 Agent Round Robin vs ARM1	128	0.0867	0.0383–0.1301	0.0%	N/A
ARM4 Agent Round Robin vs ARM1	256	0.0619	0.0113–0.1016	0.0%	N/A
ARM4 Agent Round Robin vs ARM1	512	0.0311	0.0074–0.0488	3.3%	N/A
ARM4 Agent Round Robin vs ARM1	1024	0.0103	-0.0061–0.0234	16.7%	N/A
ARM5 Hajek Weighted vs ARM1	64	0.1900	-0.0353–0.5168	13.3%	N/A
ARM5 Hajek Weighted vs ARM1	128	0.0955	-0.0189–0.2471	13.3%	N/A
ARM5 Hajek Weighted vs ARM1	256	0.0249	-0.0200–0.0944	30.0%	N/A
ARM5 Hajek Weighted vs ARM1	512	-0.0025	-0.0216–0.0087	53.3%	N/A
ARM5 Hajek Weighted vs ARM1	1024	-0.0007	-0.0156–0.0121	60.0%	N/A
ARM2 vs selected-only	64	-0.0019	-0.1053–0.1158	46.7%	N/A
ARM2 vs selected-only	128	0.0033	-0.1127–0.1522	56.7%	N/A
ARM2 vs selected-only	256	-0.0283	-0.0852–0.0459	80.0%	N/A
ARM2 vs selected-only	512	-0.0023	-0.0360–0.0378	56.7%	N/A
ARM2 vs selected-only	1024	0.0148	-0.0059–0.0355	20.0%	N/A
ARM5 vs ARM4	64	0.1122	-0.1062–0.4160	20.0%	0.1138
ARM5 vs ARM4	128	0.0088	-0.1032–0.1891	50.0%	0.0798
ARM5 vs ARM4	256	-0.0370	-0.0895–0.0463	80.0%	0.0967
ARM5 vs ARM4	512	-0.0336	-0.0446–0.0107	100.0%	0.0458
ARM5 vs ARM4	1024	-0.0111	-0.0173–0.0094	86.7%	0.0178

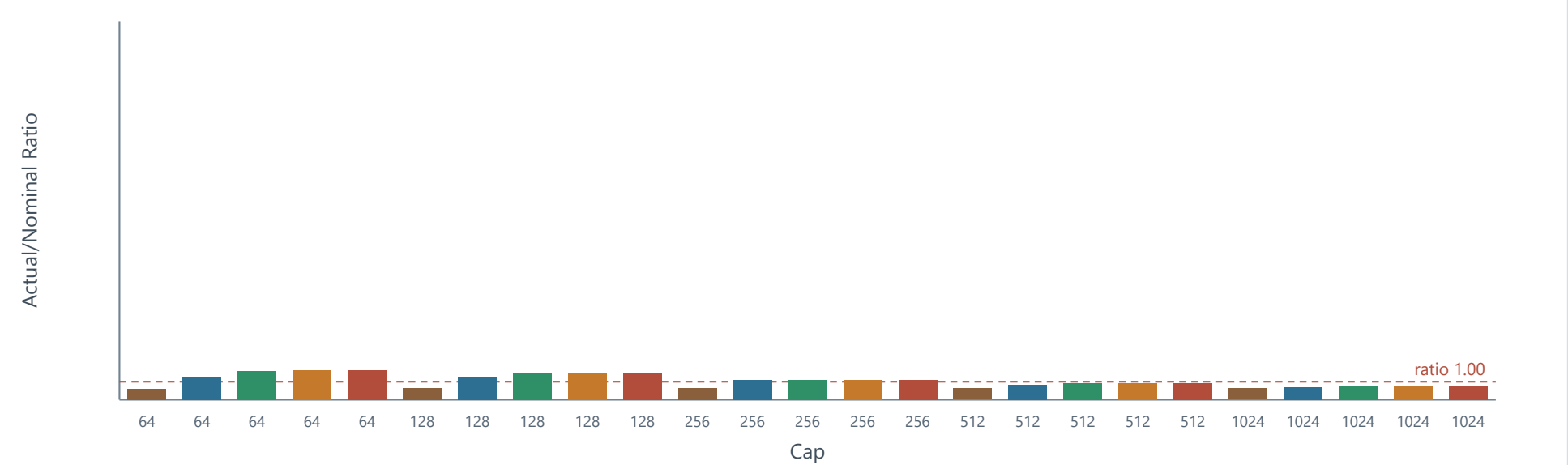
MAE median stability (static)



e. Operational cost (Token Diagnostics)

Actual vs Nominal Token Ratio by Method/Cap

Nominal uses the 15k/session planning conversion.



METHOD	CAP	NOMINAL BUDGET	ACTUAL TOKENS	ACTUAL/NOMINAL
ARM1 Global Random	64	960000	29624	3.09%
ARM2 Embedding IDW	64	960000	60667	6.32%
ARM3 Agent Round Robin Floor	64	960000	77290	8.05%
ARM4 Agent Round Robin	64	960000	78356	8.16%
ARM5 Hajek Weighted	64	960000	78356	8.16%
ARM1 Global Random	128	1920000	61171	3.19%
ARM2 Embedding IDW	128	1920000	124390	6.48%

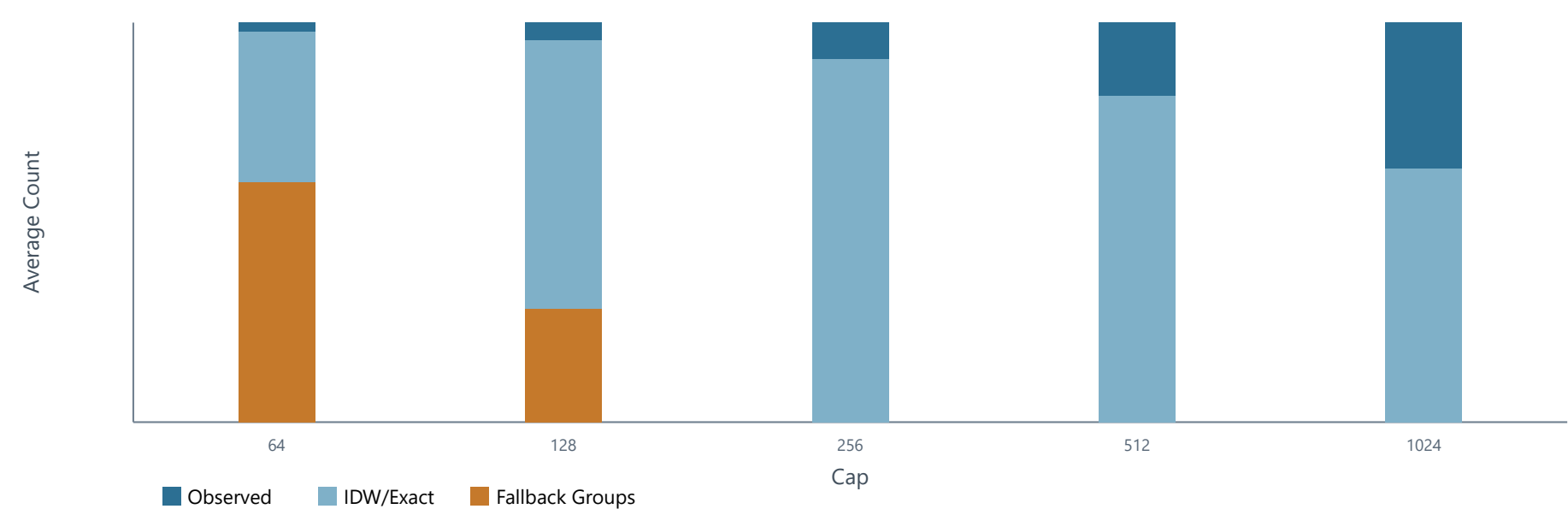
METHOD	CAP	NOMINAL BUDGET	ACTUAL TOKENS	ACTUAL/NOMINAL
ARM3 Agent Round Robin Floor	128	1920000	141096	7.35%
ARM4 Agent Round Robin	128	1920000	141397	7.36%
ARM5 Hajek Weighted	128	1920000	141397	7.36%
ARM1 Global Random	256	3840000	121259	3.16%
ARM2 Embedding IDW	256	3840000	212909	5.54%
ARM3 Agent Round Robin Floor	256	3840000	213253	5.55%
ARM4 Agent Round Robin	256	3840000	211338	5.50%
ARM5 Hajek Weighted	256	3840000	211338	5.50%
ARM1 Global Random	512	7680000	241509	3.14%
ARM2 Embedding IDW	512	7680000	323941	4.22%
ARM3 Agent Round Robin Floor	512	7680000	351070	4.57%
ARM4 Agent Round Robin	512	7680000	350753	4.57%
ARM5 Hajek Weighted	512	7680000	350753	4.57%
ARM1 Global Random	1024	15360000	482407	3.14%
ARM2 Embedding IDW	1024	15360000	543770	3.54%
ARM3 Agent Round Robin Floor	1024	15360000	569705	3.71%
ARM4 Agent Round Robin	1024	15360000	569779	3.71%
ARM5 Hajek Weighted	1024	15360000	569779	3.71%

Interpretation: Observed actual/nominal ratios span 2.42% to 8.93% with median 4.61%, SD 1.75%, and empirical p05-p95 3.07% to 8.23%. Nominal budget uses a 15k-per-session planning conversion, not a hard token pack.

f. ARM2 imputation/calibration

ARM2 IDW Provenance Stacked Composition by Cap

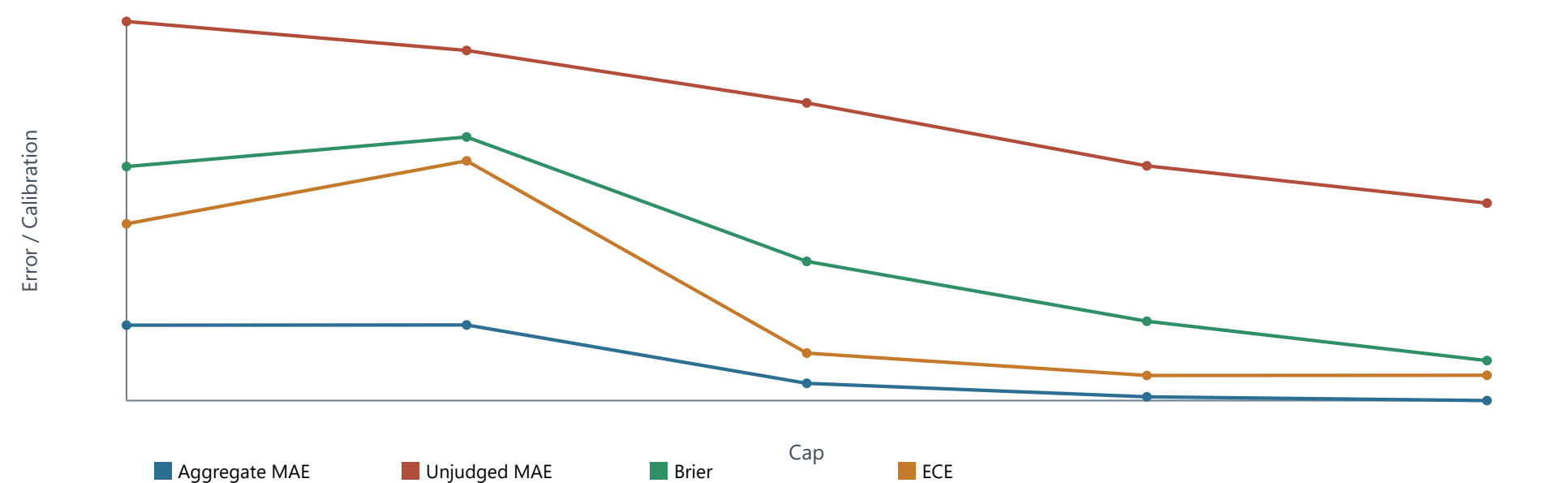
Values are per-cap averages across trials.



What this shows: Provenance composition separates observed, IDW/exact, and fallback contribution by cap.

ARM2 Quality Trends vs Cap

Trend lines are per-cap trial means.



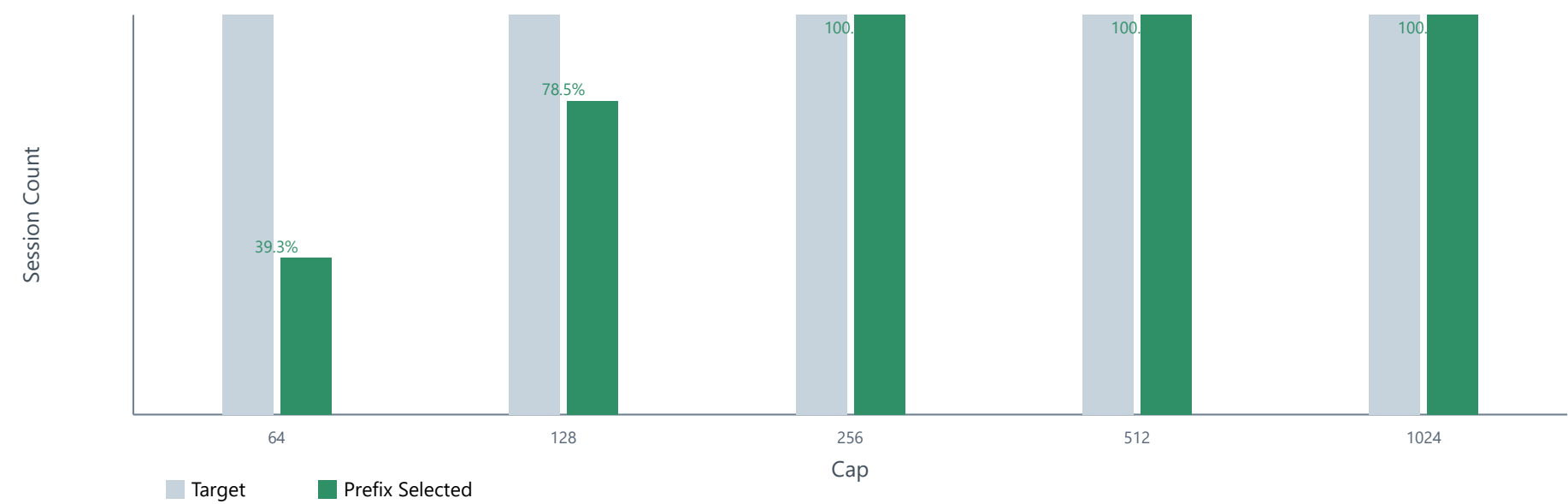
Takeaway: IDW has both better and worse outcomes across runs; use this arm for imputation/calibration analysis, not aggregate MAE leadership.

CAP	OBSERVED	IMPUTED	AGGREGATE ERROR	BRIER	ECE
64	64.0	2736.0	0.1088	0.2806	0.2186
128	128.0	2672.0	0.1091	0.3126	0.2868
256	256.0	2544.0	0.0458	0.1779	0.0786
512	512.0	2288.0	0.0312	0.1131	0.0543
1024	1024.0	1776.0	0.0271	0.0705	0.0545

g. ARM3 floor behavior and ARM4/5 relationship

ARM3 Floor Target vs Prefix Selected by Cap

Green bars show realized prefix counts; labels show completion percentage.



Takeaway: ARM3 floor mechanics explain its representation guarantees; ARM4 and ARM5 share membership while ARM5 changes only the estimator.

Membership Coverage (Print)

METHOD	CAP	SELECTED AGENTS	REPRESENTED STRATA	AGENT COVERAGE
ARM1 Global Random	64	10	36	9.56%

METHOD	CAP	SELECTED AGENTS	REPRESENTED STRATA	AGENT COVERAGE
ARM2 Embedding IDW	64	47	58	44.67%
ARM3 Agent Round Robin Floor	64	64	63	60.95%
ARM4 Agent Round Robin	64	64	63	60.95%
ARM5 Hajek Weighted	64	64	63	60.95%
ARM1 Global Random	128	16	58	14.98%
ARM2 Embedding IDW	128	81	109	76.76%
ARM3 Agent Round Robin Floor	128	105	118	100.00%
ARM4 Agent Round Robin	128	105	120	100.00%
ARM5 Hajek Weighted	128	105	120	100.00%
ARM1 Global Random	256	23	86	21.78%
ARM2 Embedding IDW	256	105	181	100.00%
ARM3 Agent Round Robin Floor	256	105	185	100.00%
ARM4 Agent Round Robin	256	105	188	100.00%
ARM5 Hajek Weighted	256	105	188	100.00%
ARM1 Global Random	512	35	126	33.17%
ARM2 Embedding IDW	512	105	222	100.00%
ARM3 Agent Round Robin Floor	512	105	256	100.00%
ARM4 Agent Round Robin	512	105	257	100.00%
ARM5 Hajek Weighted	512	105	257	100.00%
ARM1 Global Random	1024	54	177	50.95%
ARM2 Embedding IDW	1024	105	252	100.00%
ARM3 Agent Round Robin Floor	1024	105	287	100.00%
ARM4 Agent Round Robin	1024	105	287	100.00%
ARM5 Hajek Weighted	1024	105	287	100.00%

Arm3 Floor Status (Print)

METHOD	CAP	ELIGIBLE >=3	SELECTED >=3	FLOOR TARGET	FLOOR PREFIX
ARM1 Global Random	64	28	5	0	0
ARM2 Embedding IDW	64	28	4	0	0
ARM3 Agent Round Robin Floor	64	28	0	163	64
ARM4 Agent Round Robin	64	28	0	0	0
ARM5 Hajek Weighted	64	28	0	0	0
ARM1 Global Random	128	28	6	0	0
ARM2 Embedding IDW	128	28	13	0	0
ARM3 Agent Round Robin Floor	128	28	0	163	128

METHOD	CAP	ELIGIBLE >=3	SELECTED >=3	FLOOR TARGET	FLOOR PREFIX
ARM4 Agent Round Robin	128	28	0	0	0
ARM5 Hajek Weighted	128	28	0	0	0
ARM1 Global Random	256	28	7	0	0
ARM2 Embedding IDW	256	28	26	0	0
ARM3 Agent Round Robin Floor	256	28	28	163	163
ARM4 Agent Round Robin	256	28	28	0	0
ARM5 Hajek Weighted	256	28	28	0	0
ARM1 Global Random	512	28	9	0	0
ARM2 Embedding IDW	512	28	26	0	0
ARM3 Agent Round Robin Floor	512	28	28	163	163
ARM4 Agent Round Robin	512	28	28	0	0
ARM5 Hajek Weighted	512	28	28	0	0
ARM1 Global Random	1024	28	12	0	0
ARM2 Embedding IDW	1024	28	26	0	0
ARM3 Agent Round Robin Floor	1024	28	28	163	163
ARM4 Agent Round Robin	1024	28	28	0	0
ARM5 Hajek Weighted	1024	28	28	0	0

Five Largest Agents: Sentinel Drilldown

Five dense 500-session agents only (not all-agent validation)

METHOD	CAP	AGENT	SELECTED/CENSUS RATE	ABS ERROR	N
ARM1 Global Random	64	tenant-a365-deterministic finance-operations-agent	80.32% / 82.00%	0.0778	11.4
ARM1 Global Random	64	tenant-a365-deterministic procurement-agent	72.93% / 76.00%	0.0990	11.6
ARM1 Global Random	64	tenant-a365-deterministic travel-coordinator-agent	70.05% / 71.00%	0.0971	12.5
ARM1 Global Random	64	tenant-a365-deterministic software-delivery-agent	69.37% / 64.00%	0.1127	10.8
ARM1 Global Random	64	tenant-a365-deterministic it-support-agent	66.64% / 68.00%	0.1154	11.5
ARM2 Embedding IDW	64	tenant-a365-deterministic procurement-agent	91.67% / 76.00%	0.2833	0.4
ARM2 Embedding IDW	64	tenant-a365-deterministic finance-operations-agent	85.71% / 82.00%	0.2714	0.2

METHOD	CAP	AGENT	SELECTED/CENSUS RATE	ABS ERROR	N
ARM2 Embedding IDW	64	tenant-a365-deterministic travel-coordinator-agent	62.50% / 71.00%	0.4475	0.3
ARM2 Embedding IDW	64	tenant-a365-deterministic it-support-agent	60.00% / 68.00%	0.4640	0.5
ARM2 Embedding IDW	64	tenant-a365-deterministic software-delivery-agent	58.33% / 64.00%	0.4767	0.4
ARM3 Agent Round Robin Floor	64	tenant-a365-deterministic software-delivery-agent	84.00% / 64.00%	0.4048	0.8
ARM3 Agent Round Robin Floor	64	tenant-a365-deterministic procurement-agent	83.33% / 76.00%	0.3267	0.6
ARM3 Agent Round Robin Floor	64	tenant-a365-deterministic finance-operations-agent	78.95% / 82.00%	0.3147	0.6
ARM3 Agent Round Robin Floor	64	tenant-a365-deterministic travel-coordinator-agent	77.27% / 71.00%	0.3855	0.7
ARM3 Agent Round Robin Floor	64	tenant-a365-deterministic it-support-agent	63.16% / 68.00%	0.4526	0.6
ARM4 Agent Round Robin	64	tenant-a365-deterministic travel-coordinator-agent	87.50% / 71.00%	0.3425	0.5
ARM4 Agent Round Robin	64	tenant-a365-deterministic finance-operations-agent	83.33% / 82.00%	0.2867	0.6
ARM4 Agent Round Robin	64	tenant-a365-deterministic procurement-agent	72.73% / 76.00%	0.3818	0.4
ARM4 Agent Round Robin	64	tenant-a365-deterministic it-support-agent	71.43% / 68.00%	0.4229	0.7
ARM4 Agent Round Robin	64	tenant-a365-deterministic software-delivery-agent	60.00% / 64.00%	0.4720	0.7
ARM5 Hajek Weighted	64	tenant-a365-deterministic travel-coordinator-agent	87.50% / 71.00%	0.3425	0.5
ARM5 Hajek Weighted	64	tenant-a365-deterministic finance-operations-agent	83.33% / 82.00%	0.2867	0.6
ARM5 Hajek Weighted	64	tenant-a365-deterministic procurement-agent	72.73% / 76.00%	0.3818	0.4
ARM5 Hajek Weighted	64	tenant-a365-deterministic it-support-agent	71.43% / 68.00%	0.4229	0.7
ARM5 Hajek Weighted	64	tenant-a365-deterministic software-delivery-agent	60.00% / 64.00%	0.4720	0.7
ARM1 Global Random	128	tenant-a365-deterministic finance-operations-agent	83.84% / 82.00%	0.0710	21.5

METHOD	CAP	AGENT	SELECTED/CENSUS RATE	ABS ERROR	N
ARM1 Global Random	128	tenant-a365-deterministic procurement-agent	74.49% / 76.00%	0.0551	24.2
ARM1 Global Random	128	tenant-a365-deterministic travel-coordinator-agent	68.56% / 71.00%	0.0608	21.6
ARM1 Global Random	128	tenant-a365-deterministic it-support-agent	68.10% / 68.00%	0.0822	23.3
ARM1 Global Random	128	tenant-a365-deterministic software-delivery-agent	66.15% / 64.00%	0.0634	22.6
ARM2 Embedding IDW	128	tenant-a365-deterministic finance-operations-agent	93.33% / 82.00%	0.2227	0.5
ARM2 Embedding IDW	128	tenant-a365-deterministic procurement-agent	89.47% / 76.00%	0.2947	0.6
ARM2 Embedding IDW	128	tenant-a365-deterministic travel-coordinator-agent	76.00% / 71.00%	0.3908	0.8
ARM2 Embedding IDW	128	tenant-a365-deterministic it-support-agent	68.18% / 68.00%	0.4345	0.7
ARM2 Embedding IDW	128	tenant-a365-deterministic software-delivery-agent	52.17% / 64.00%	0.4939	0.8
ARM3 Agent Round Robin Floor	128	tenant-a365-deterministic finance-operations-agent	85.00% / 82.00%	0.2340	1.8
ARM3 Agent Round Robin Floor	128	tenant-a365-deterministic it-support-agent	70.00% / 68.00%	0.3000	1.8
ARM3 Agent Round Robin Floor	128	tenant-a365-deterministic procurement-agent	70.00% / 76.00%	0.3320	1.8
ARM3 Agent Round Robin Floor	128	tenant-a365-deterministic travel-coordinator-agent	68.33% / 71.00%	0.2973	1.8
ARM3 Agent Round Robin Floor	128	tenant-a365-deterministic software-delivery-agent	58.33% / 64.00%	0.3207	1.9

Top-Five Agent Absolute Error Heatmap by Method/Cap

Cell values are mean absolute error across top-five rows.

	cap 64	cap 128	cap 256	cap 512	cap 1024
ARM1 Global Random	0.100	0.066	0.054	0.037	0.021
ARM2 Embedding IDW	0.389	0.367	0.098	0.042	0.025
ARM3 Agent Round R	0.377	0.297	0.107	0.033	0.018
ARM4 Agent Round R	0.381	0.308	0.097	0.033	0.018
ARM5 Hajek Weighte	0.381	0.308	0.097	0.033	0.018

Observation: At low caps this sentinel panel shows n=0/1 volatility and missing selections; at higher caps selected rates converge toward census rates.

Limitations and Validity

Scope and caveats for interpretation

- Synthetic sessions and labels; this is not production-ground-truth evidence.

- Fixed population and deterministic replays; no p-values/generalization inference.
- Classification ambiguity is high; use-case coverage is provisional.
- ARM2 includes model-assisted IDW assumptions and fallback behavior.
- ARM5 weighting can be unstable at low caps.
- 15k planning conversion is conservative and approximate.
- Sentinel drilldown covers only five largest dense agents.

Recommendations and Next Steps

Actionable next work and update triggers

Accuracy-first: ARM1 at caps 64/128/256.

Representation-first: ARM4/ARM5, with explicit provisional use-case caveat.

Balanced objective: ARM5 at 512; ARM5 vs ARM1 at 1024 is a near-tie due to mean-median disagreement.

1. Run a production shadow evaluation with matched cap/seed structure.
2. Perform a labeled Maven classification audit.
3. Calibrate nominal token conversion empirically from production traces.
4. Increase trials only if rank ordering remains ambiguous.

Evidence that would change this decision: shadow-run rank reversal, improved classification confidence profile, or major token calibration shift.

Appendix

Detailed taxonomy, trial diagnostics, and methodology references

Top Use Cases (GUID)

- fc8e4c9e-d911-4fa8-8719-708f9b2b9651: 260
- 2c698729-251a-4ebe-aec5-c81e2be1dce9: 218
- 0231ac82-2e61-4c02-907e-17ee61eefd56: 136

- 51f99f1a-4357-4cf4-9777-c353fc24d48d: 125
- 61d98c2f-1954-4e3d-95cf-d1f0c79fab59: 117
- ab3f7996-8a78-43e5-b3e3-bbba78b2667d: 113
- 4f387cbf-1796-4201-8c45-99dde6309f87: 104
- d4fc97da-3222-4678-a9e5-b075087da334: 85
- 6ce3bd74-e441-4886-b247-a0e4befabbcd: 64
- fd2ccf35-9332-4cb7-85f7-7857e943a4e8: 56

Total unique: 311

Explicitly unclassified: 0

Below-0.30 fallback: 23

Top Sub-subcategory

- Financial Compliance Review: 260
- Travel Policy Compliance: 218
- Incomplete or Vague Prompt: 136
- Issue Resolution: 125
- Rollback Procedures: 118
- Dependency Risk: 117
- Third-Party Risk: 104
- Compliance Screening: 85
- Graceful Degradation: 64
- Service Request Handling: 56

Total unique: 306

Confidence / Status

Confidence Levels

- 1: 2564
- 2: 132

- 3: 81
- undetermined: 23

Status

- Ambiguous: 2569
- Corroborated: 146
- Agree: 85

Detailed IDW, membership, taxonomy, and methodology material is printed in the corresponding sections above; it is not repeated here.